

Lunar Formation

via Triple Phase Transition

in the Differentiating Proto-Earth

Michel Debailleul

Geophysicist — Université libre de Bruxelles (ULB)

ORCID: [0009-0003-1222-1433](https://orcid.org/0009-0003-1222-1433)

michel.debailleul@yahoo.fr

2026

© 2026 Michel Debailleul — Licence CC BY-NC 4.0



Abstract. This work proposes a theory of lunar formation grounded in a thermodynamic necessity: any Earth-mass planet emerges from accretion in a state of near-total silicate mantle melting. Accretion energy exceeds total fusion energy by a factor of 155 — the proto-Earth could only have been a rapidly rotating magma sphere immersed in a Hadean inferno: a raging T-Tauri Sun, a silicate atmosphere at 2 000–4 000 K, incessant planetesimal bombardment, still-active short-lived radionuclides, and a chaotic axial obliquity between 40° and 70°.

It is in this context — and nowhere else — that the **Triple Phase Transition (TPT)** is triggered. Its unique driver is the **progressive Fe-Ni segregation**, which simultaneously governs three regime changes: **(1)** a rheological transition — the silicate melt acquires a yield stress (Bingham-Herschel law) and organises a coherent toroidal flow in the intertropical belt $|\varphi| < 30^\circ$, the Coherent Magmatic Torus (CMT); **(2)** a mechanical transition — the CMT crosses the instability threshold and produces two to three episodes of cohesive ejection beyond the Roche radius; **(3)** a magnetic transition — the terrestrial dynamo emerges ≈ 300 Myr after the ejections cease.

The central equation of the theory: $M_{\text{ej}}^{\text{max}} \approx 2\Omega \sin \varepsilon \cdot U_{\text{crit}} \cdot M_{\text{CMT}}/g_{\text{eff}}(0) \approx 2.2\text{--}4.0 \times 10^{22}$ kg per event. With $N = 2\text{--}3$ and a capture efficiency $f_{\text{cap}} \approx 0.70$, the lunar mass is recovered. Each episode leaves a detectable **stratigraphic signature**: seismic interface between 200 and 315 km depth, Fe/Si gradient at depth, near-side/far-side crustal dichotomy. The **Chang’e 7** mission (seismometer, South Pole, August 2026) constitutes the nearest validation window.

Epistemic note. This work submits a theory, not certainties. All hypotheses are ranked, all limitations are acknowledged, the validation programme is defined and dated.

Contents

1	An Active Inferno: the Hadean Context	3
1.1	The T-Tauri Sun: a Star in Eruption	3
1.2	The Silicate Atmosphere: a Thermal Cocoon at 2 000–4 000 K	3
1.3	Continuous Planetary Bombardment	4
1.4	Short-Lived Radioactivity: an Internal Heat Source	4
1.5	Hadean Obliquity: 40°–70°, a Central Parameter	4
2	The Initial State of the Proto-Earth	5
2.1	A Deeply Molten Mantle: Thermodynamic Necessity	5
2.2	Initial Rotation: $T_{\text{rot}}^{(0)} \approx 5$ h, a Constrained Value	6
2.3	Maclaurin Geometry: Cradle of the CMT	6
3	The Intertropical Belt $\varphi < 30^\circ$: the Ejection Site	7
4	The CMT as a Pure Toroidal Flow: Response to the Taylor-Proudman Objection	8
4.1	Poloidal/Toroidal Decomposition	8
4.2	Why Curved Columns Do Not Extend Beyond the Active Band	8
4.3	Response to the Objection	9
5	The Three Coupled Transitions	9
5.1	Transition 1 — Rheological: the Bingham-Herschel Law	9
5.2	Transition 2 — Mechanical: Instability and Ejection	10
5.2.1	Derivation of λ from the Equations of Motion	10
5.2.2	The Double Potential Well: Episodes as Natural Bifurcations	12
5.2.3	The Central Equation: Maximum Ejectable Mass	13
5.2.4	Mass Budget: N=2–3 Episodes + SPA Impactor	13
5.2.5	Transient Post-Ejection Precession	14
5.3	Transition 3 — Magnetic: the Hadean Dynamo	14
6	Chronology and Observational Constraints	16
6.1	Proposed Chronology	16

6.2	The Three Observational Constraints	16
7	Crustal Dichotomy: Natural Consequence of the Asymmetric Second Episode	17
8	Falsifiable Predictions	18
9	Upcoming Missions: Three Validation Windows	19
9.1	Chang’e 7: Imminent Seismic Validation	19
9.2	The South Pole-Aitken Basin: a Stratigraphic Revealer, not merely a Crater	19
9.3	Artemis III: Landing on a Stratigraphic Treasure	20
10	Discussion	22
10.1	Comparison with the Giant Impact Model	22
10.2	Acknowledged Limitations	22
10.3	Validation Programme	23
11	Conclusion	24
A	Adiabatic Temperature Profile (Supplement to H0)	24
B	Dynamo Delay Estimate	24
C	Interactive Numerical Simulation	25

1 An Active Inferno: the Hadean Context

Before entering the physics of the Triple Phase Transition, it is essential to understand the setting in which it occurs. This setting is not a quiet void: it is one of the most violent periods in the history of the Solar System.

1.1 The T-Tauri Sun: a Star in Eruption

In its first few million years, the Sun was in its T-Tauri phase. In practice: stellar winds 100 to 1 000 times more intense than today (Feigelson & Montmerle, 1999; Airapetian et al., 2016), coronal mass ejections (CMEs) violently injecting energy into the proto-Earth’s atmosphere, extreme X-ray and UV radiation maintaining the surface above 3 000 K. This Sun is not a background detail: it is an active physical agent that extends the TPT activation window well beyond what the accretion energy budget alone would predict.

T-Tauri outbursts are also natural triggers for CMT instability events: an energetic external perturbation can push the system over the potential barrier separating the confined state from the ejective state (Section 5.2.2).

1.2 The Silicate Atmosphere: a Thermal Cocoon at 2 000–4 000 K

At $t = 0$, the proto-Earth is enveloped in a dense silicate atmosphere — rock vapour, SiO, gaseous Mg and Fe — maintained between 2 000 and 4 000 K. This cocoon plays three distinct physical roles:

Thermal insulation. It blocks radiative surface cooling, keeping the mantle molten well beyond the accretion energy budget alone.

Confinement of the CMT flux. Its pressure, $\sim 10^5\text{--}10^7$ Pa, opposes lateral expansion of the flux during ejection and contributes to the cohesion of the sheet.

High sound speed. At $T_{\text{atm}} \approx 3\,000$ K, the sound speed in the silicate atmosphere reaches $c_{\text{sound}} \approx 1\,500\text{ m s}^{-1}$, yielding an external Mach number $\text{Ma}_{\text{ext}} \approx 6\text{--}8$: the flux is hypersonic, yet less disruptive than the initial $\text{Mach} \approx 11.5$ — cohesion is better preserved in this realistic regime.

1.3 Continuous Planetary Bombardment

The protoplanetary disc is still populated with planetesimals and planetary embryos (Agnor et al., 1999; Kokubo & Genda, 2010; Chambers, 2001). These impactors act on three fronts: they re-inject thermal energy into the mantle (sustaining melting), they stochastically perturb rotation and obliquity, and they fracture any nascent proto-crust before it can consolidate. The Nice model (Tsiganis et al., 2005) demonstrates that giant-planet orbital instability amplifies this bombardment over the first several hundred million years.

1.4 Short-Lived Radioactivity: an Internal Heat Source

The isotopes ^{26}Al (half-life: 0.72 Myr) and ^{60}Fe (half-life: 2.6 Myr) are still active in the first few million years (Huss et al., 2009). Their decay injects an additional $\sim 3 \times 10^{29}$ J into the mantle, contributing to sustained melting throughout the entire active TPT window.

1.5 Hadean Obliquity: 40° – 70° , a Central Parameter

In the absence of any stabilising satellite, the proto-Earth’s obliquity is subject to spin-orbit instabilities. Simulations by Laskar et al. (1993a); Laskar (1993b) demonstrate that it evolves chaotically within the range $\varepsilon \in [40^\circ, 70^\circ]$, with a central value $\varepsilon \approx 55^\circ$. This parameter is decisive for the theory. The Coriolis force decomposes into a radial component (outward, destabilising) and a horizontal component:

$$f_{\text{rad}} = 2 \Omega \sin \varepsilon \cdot U, \quad f_{\text{horiz}} = 2 \Omega \cos \varepsilon \cdot U. \quad (1)$$

At $\varepsilon = 55^\circ$: $f_{\text{rad}}/f_{\text{tot}} = \sin 55^\circ \approx 0.819$, meaning 82 % of the total force is directed towards ejection. Over the full Hadean range $[40^\circ, 70^\circ]$, $\sin \varepsilon$ remains above 0.64: the mechanism operates throughout, with an optimum at 70° and a minimum at 40° .

The geometric threshold $\varepsilon = 45^\circ$ ($\sin 45^\circ = 0.707$) separates the moderately unstable regime from the strongly unstable one. The Hadean range guarantees that the proto-Earth regularly exceeds this threshold.

2 The Initial State of the Proto-Earth

2.1 A Deeply Molten Mantle: Thermodynamic Necessity

The formation of a rocky planet releases enormous gravitational energy. Physically: all matter that fell towards the centre during accretion converted its potential energy into heat. The calculation, in the homogeneous-sphere approximation (Safronov, 1969):

$$E_{\text{grav}} = \frac{3 G M_{\oplus}^2}{5 R_{\oplus}} \approx 2.49 \times 10^{32} \text{ J}. \quad (2)$$

Compared with the energy required to melt the entire silicate mantle (Stixrude & Lithgow-Bertelloni, 2014):

$$E_{\text{fus}} \approx M_{\text{mantle}} \times L_{\text{fus}} \approx 4 \times 10^{24} \times 4 \times 10^5 \approx 1.6 \times 10^{30} \text{ J}. \quad (3)$$

Established result

Energy ratio: $E_{\text{grav}}/E_{\text{fus}} \approx 155$. Accretion energy exceeds total fusion energy by two orders of magnitude (Solomatov, 2000; Elkins-Tanton, 2012; Rubie et al., 2015). The question is not *how to melt* the mantle, but *how and how fast to cool it*. Near-total melting is not a hypothesis: it is a necessary thermodynamic consequence.

Added to this budget: Fe-Ni segregation heat ($\approx 3 \times 10^{30}$ J, Flasar & Birch 1973; Rubie et al. 2003), short-lived radioactivity, and the T-Tauri Sun. The proto-Earth is guaranteed to be in deep, prolonged melting.

Central hypothesis

Hypothesis H0 — Initial state. At $t = 0$, the silicate mantle of the proto-Earth is in near-total melting — whether one calls this wholesale fusion or a very-deep magma ocean, the CMT physics is identical in both cases: no stable solid interface exists during the active window. Fe-Ni is dissolved in the molten silicate, with no differentiated core. Four complementary arguments document this state (Melosh, 1990; Rubie et al., 2015; Johansen & Lambrechts, 2017; Nguyen et al., 2020).

2.2 Initial Rotation: $T_{\text{rot}}^{(0)} \approx 5 \text{ h}$, a Constrained Value

The rotation period of the proto-Earth at $t = 0$ is not a free parameter: it is constrained by conservation of the total angular momentum of the Earth-Moon system. The argument is as follows: the total angular momentum measured today ($L_{\text{total}} \approx 3.46 \times 10^{34} \text{ J s}$) has been conserved since formation. Tracing back tidal migration, one extracts the required initial spin:

$$L_{\text{spin}}^{(0)} \approx 3.02 \times 10^{34} \text{ J s}.$$

The moment-of-inertia factor of the undifferentiated proto-Earth (Fe-Ni dispersed in molten silicate, hence close to a homogeneous sphere) is $k_{\text{proto}} \approx 0.378$, giving:

$$T_{\text{rot}}^{(0)} = \frac{2\pi I_{\text{proto}}}{L_{\text{spin}}^{(0)}} = \frac{2\pi \times 9.68 \times 10^{37}}{3.02 \times 10^{34}} \approx 5 \text{ h}. \quad (4)$$

This value of 5 h is consistent with the 4–6 h range from N-body accretion simulations (Agnor et al., 1999; Kokubo & Genda, 2010): rapid rotation is the norm at the end of accretion, not an exception. The Coriolis force is maximal at $t = 0$, before any tidal braking. This window does not recur.

2.3 Maclaurin Geometry: Cradle of the CMT

A fully molten rotating body takes the shape of a Maclaurin spheroid. For $T_{\text{rot}} = 5 \text{ h}$, the rotation parameter $m = \Omega^2 R_{\oplus}^3 / (GM_{\oplus}) \approx 0.096$ yields: eccentricity $e \approx 0.502$, flattening ≈ 0.130 , intertropical bulge $\delta R \approx 895 \text{ km}$.

Physically: effective gravity is minimal in the oblique intertropical band $|\varphi| < 30^\circ$.

Molten matter preferentially accumulates in this bulge — this is the natural, geometrically constrained cradle of the CMT. At $T_{\text{rot}} = 3.5 \text{ h}$, δR reaches 1 484 km; at 3.0 h, 2 191 km: the intertropical band is even more pronounced at faster rotation.

3 The Intertropical Belt $|\varphi| < 30^\circ$: the Ejection Site

A fundamental point, often misunderstood: the CMT is **not an equatorial structure**. It is a coherent zonal band extending across $\pm 30^\circ$ of oblique latitude — roughly one third of the proto-Earth’s surface. This extent results from three independent mechanisms that converge.

(1) Maclaurin geometry. Effective gravity is minimal in the band $|\varphi| < 30^\circ$. Molten matter accumulates there by centrifugal effect, independently of any other argument. This is a consequence of the ellipsoidal shape imposed by rapid rotation.

(2) Geostrophic inverse cascade. The Rossby number of the CMT is deeply geostrophic:

$$\text{Ro}_{\text{CMT}} = \frac{U}{2\Omega R_\oplus} \approx \frac{9.5 \times 10^3}{2 \times 3.49 \times 10^{-4} \times 6.371 \times 10^6} \approx 2.1 \times 10^{-3} \ll 1. \quad (5)$$

In this regime, turbulent energy cascades toward large scales (inverse cascade). The Rhines scale:

$$L_\beta = \sqrt{\frac{U}{\beta}}, \quad \beta = \frac{2\Omega \cos \varepsilon}{R_\oplus}, \quad (6)$$

gives $L_\beta \approx 0.52 R_\oplus$ — corresponding to harmonic mode $\ell = 1$, $m = 1$: the largest coherent structure possible in a sphere. The CMT is the statistical attractor of this cascade (Rhines, 1975; Sukoriansky et al., 2007).

(3) Radial Coriolis force. $f_{\text{rad}} = 2\Omega U \sin \varepsilon$ concentrates the flux in the oblique intertropical band and keeps it coherent there.

These three mechanisms are *independent* and *convergent* toward the same band $|\varphi| < 30^\circ$. Their simultaneous convergence is an argument for the structural robustness of the theory.

4 The CMT as a Pure Toroidal Flow: Response to the Taylor-Proudman Objection

Any geophysical fluid physicist confronted with the CMT will naturally raise the following objection: in a rapidly rotating fluid ($\text{Ro} \ll 1$), the Taylor-Proudman theorem imposes $(\boldsymbol{\Omega} \cdot \nabla)\mathbf{u} = 0$, which produces columnar structures parallel to $\boldsymbol{\Omega}$, not a single oblique band $|\varphi| < 30^\circ$. The objection is legitimate and deserves a precise answer.

4.1 Poloidal/Toroidal Decomposition

Any velocity field in a spherical shell decomposes into a **poloidal** part (radial and meridional components) and a **toroidal** part (pure azimuthal component):

$$\mathbf{u} = \mathbf{u}_{\text{pol}} + \mathbf{u}_{\text{tor}}, \quad \mathbf{u}_{\text{tor}} = \nabla \times (\psi \hat{\mathbf{r}}).$$

The Taylor-Proudman theorem, in its classical form, constrains the poloidal field (which transports vorticity along $\boldsymbol{\Omega}$) and produces Taylor columns. It does **not** constrain the pure toroidal field.

For a purely azimuthal flow independent of longitude ($u_r = u_\theta = 0$ on average, $u_\varphi = u_\varphi(r, \varphi)$), the Taylor-Proudman constraint is automatically satisfied:

$$(\boldsymbol{\Omega} \cdot \nabla) \mathbf{u}_{\text{tor}} = 0 \tag{7}$$

for any toroidal field, without additional condition. The **balance is cyclostrophic**, not geostrophic: the centrifugal pressure gradient balances the Coriolis force in the azimuthal plane.

4.2 Why Curved Columns Do Not Extend Beyond the Active Band

In a spherical shell rotating obliquely ($\varepsilon \neq 0$), Taylor columns are curved along field lines of projected $\boldsymbol{\Omega}$. They tend to channel flow away from their source — unless a cut-off mechanism stops them.

This is precisely the role of the yield stress τ_y . In a Bingham-Herschel fluid, flow propagates only in regions where the stress exceeds τ_y . Once columns penetrate a

region where $\tau < \tau_y$, they are truncated: the magma remains frozen there (Frigaard & Nouar, 2017; Balmforth & Craster, 2001). The band $|\varphi| < 30^\circ$ is precisely the region where effective gravity is minimal (Maclaurin) and Coriolis stress is maximal ($\varepsilon \approx 55^\circ$): this is the only region where $\tau > \tau_y$. Curved columns emanating from the active band are thus naturally truncated at its boundaries.

4.3 Response to the Objection

The CMT is a pure toroidal flow. The Taylor-Proudman objection does not apply to it. What remains to be demonstrated numerically — identified as Priority 2 in this work — is that a *single* torus forms in the band $|\varphi| < 30^\circ$ rather than several alternating bands. This is a question of simulation, not of theoretical inconsistency.

5 The Three Coupled Transitions

The unique driver is the progressive Fe-Ni segregation. It simultaneously governs three regime changes: the rheological transition opens the window, the mechanical transition produces the ejection episodes, and the magnetic transition closes the loop with the terrestrial dynamo.

5.1 Transition 1 — Rheological: the Bingham-Herschel Law

Phase transition — Rheological — Bingham-Herschel

A Newtonian fluid flows as soon as any stress is applied. A Bingham-Herschel fluid, by contrast, **resists like a solid** until the applied stress exceeds a threshold τ_y . It is this threshold that allows the CMT to accumulate energy without dispersing, and to eject as a single cohesive mass when the mechanical threshold is crossed.

The Bingham-Herschel law (Bingham, 1922; Herschel & Bulkley, 1926):

$$\tau = \tau_y + k \dot{\gamma}^n \quad \text{for } \tau > \tau_y. \quad (8)$$

For silicate melt at 3 000–3 500 K with solid fraction $\phi \approx 0.1\text{--}0.3$: $\tau_y \in [10^2, 10^4]$ Pa (Giordano et al., 2008), $n = 1.2$ (Caricchi et al., 2007).

The yield stress τ_y depends exponentially on temperature:

$$\tau_y(T) \approx \tau_y^{(0)} \exp\left(-\frac{E_a}{RT}\right), \quad E_a \approx 150\text{--}250 \text{ kJ mol}^{-1}. \quad (9)$$

At $T = 3\,000 \text{ K}$: $\tau_y \approx 10^2 \text{ Pa}$ — the flux flows easily. At $T^* \approx 1\,800 \text{ K}$: $\tau_y \sim 10^6 \text{ Pa}$ — cohesive ejection becomes physically inaccessible.

$T^* \approx 1\,800 \text{ K}$ is therefore the **thermal lock** of the theory: when the mantle cools below this threshold, the first transition closes irreversibly. The Arrhenius law thus transforms progressive cooling into a natural extinction mechanism for the TPT window — no ad hoc condition, a thermodynamic consequence.

'A $\bar{a}$ structure of the CMT flux. The toroidal flow has a two-phase architecture: a **mobile plastic core** at high temperature, surrounded by a **fragmented clinker crust**. This is not a problem for cohesion: it is its guarantor. The outer crust protects the core against Kelvin-Helmholtz instabilities at the flux/atmosphere interface. A quench-glass envelope (analogous to the glassy margins of pillow lavas (Hekinian et al., 1989)), formed in $\approx 7 \text{ min}$ on contact with the atmosphere, completes this rheological shielding. The effective cohesion parameter $\Pi_{\text{eff}} \approx 3.6 \times 10^8 \gg 1$ confirms that fragmentation is suppressed at all relevant scales.

5.2 Transition 2 — Mechanical: Instability and Ejection

Phase transition — Mechanical — CMT Flux Instability

The CMT accumulates energy in the intertropical belt. When the centrifugal and Coriolis force exceeds the confining gravity, the stability coefficient λ changes sign: the flux transitions from the confined state to the ejective state. This is the second phase transition — mechanical.

5.2.1 Derivation of λ from the Equations of Motion

Consider a CMT fluid particle with azimuthal velocity U at the proto-Earth's surface ($r = R_\oplus$), in the frame rotating at Ω . The radial equation of motion reads:

$$\rho \ddot{r} = -\rho g_{\text{eff}}(0) + \rho \cdot 2\Omega \sin \varepsilon \cdot U + \rho \frac{U^2(\alpha - 1)}{R_\oplus},$$

where $g_{\text{eff}}(0) = g_0 - \Omega^2 R_{\oplus}$ is the effective gravity in the intertropical band, and α is the rotation-profile exponent ($\alpha > 1$: super-rigid profile). Setting $\delta r \propto e^{\lambda t}$, instability requires $\lambda > 0$, with:

$$\lambda(U, \varepsilon, \alpha) = \underbrace{-\frac{g_{\text{eff}}(0)}{R_{\oplus}}}_{\lambda_1 < 0} + \underbrace{\frac{2\Omega \sin \varepsilon}{R_{\oplus}} U}_{\lambda_2 > 0} + \underbrace{\frac{(\alpha - 1)}{R_{\oplus}^2} U^2}_{\lambda_3 \geq 0} \quad (10)$$

Physically: λ_1 is the restoring gravity (stabilising), λ_2 is the radial component of the Coriolis force (destabilising, increases with ε), λ_3 is the angular momentum gradient term (destabilising if $\alpha > 1$).

Numerical values at $T_{\text{rot}} = 5 \text{ h}$, $\varepsilon = 55^\circ$, $\alpha = 1.15$, $U = U_{\text{crit}} = 10.73 \text{ km s}^{-1}$:

$$\begin{aligned} \lambda_1 &= -9.034 / (6.371 \times 10^6) = -1.418 \times 10^{-6} \text{ s}^{-2}, \\ \lambda_2 &= +9.63 \times 10^{-7} \text{ s}^{-2}, \\ \lambda_3 &= +4.25 \times 10^{-7} \text{ s}^{-2}, \\ \lambda &\approx -2.95 \times 10^{-8} \text{ s}^{-2}. \end{aligned} \quad (11)$$

The system is *marginally stable* at $\varepsilon = 55^\circ$: near the threshold but not yet unstable. The instability window ($\lambda > 0$) opens at $\varepsilon \gtrsim 58^\circ$, as shown in Table 1. Note that neither λ_2 nor λ_3 alone suffices: their conjunction is required, which constitutes a falsifiable prediction ($\alpha > 1$ is required).

Table 1: Computed instability threshold as a function of obliquity ($\Omega_0 = 3.491 \times 10^{-4} \text{ rad s}^{-1}$, $\alpha = 1.15$, $U_{\text{crit}} = 10.73 \text{ km s}^{-1}$). The Hadean range $[40^\circ, 70^\circ]$ spans the stable/unstable transition.

ε	$U_{\text{thresh}} \text{ (km s}^{-1}\text{)}$	U_t/U_{crit}	Regime
40°	12.25	1.14	stable
45°	11.73	1.09	stable
50°	11.29	1.05	stable
55°	10.90	1.016	stable (very close to threshold; $\lambda = 0$ exact at $\varepsilon \approx 58.2^\circ$)
60°	10.58	0.986	unstable ($\lambda > 0$)
65°	10.32	0.962	unstable
70°	10.11	0.942	unstable

This table constitutes **quantified prediction P5**: the active instability window is $\varepsilon \in [58^\circ, 70^\circ]$. Over the chaotic Hadean distribution $[40^\circ, 70^\circ]$, $\approx 40\%$ of configurations are unstable: the $N = 2-3$ episodes occur during phases when ε exceeds 58° , consistent with the chaotic dynamics of Laskar et al. (1993a); Laskar (1993b).

5.2.2 The Double Potential Well: Episodes as Natural Bifurcations

Integrating $\lambda(U) = -dV/dU$, one obtains the effective potential of the CMT:

$$V(U) = \frac{|g_{\text{eff}}|}{R_\oplus} U - \frac{\Omega \sin \varepsilon}{R_\oplus} U^2 - \frac{(1 - \alpha)}{3 R_\oplus^2} U^3 + V_0. \quad (12)$$

This potential exhibits a **double-well structure**: two stable states separated by a barrier ΔE_{barr} .

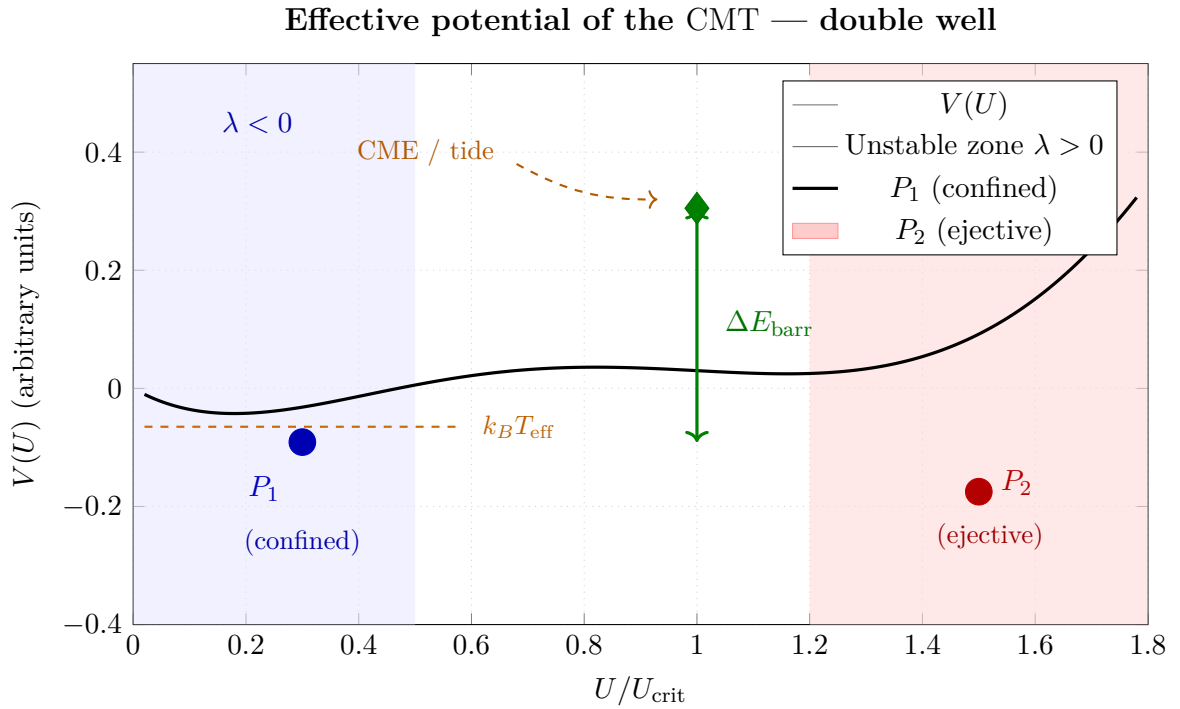


Figure 1: Effective potential $V(U)$ of the CMT as a function of normalised azimuthal velocity U/U_{crit} . P_1 : confined well — the CMT accumulates energy. P_2 : ejective well — cohesive ejection is favoured. The barrier ΔE_{barr} can be crossed spontaneously (by progressive accumulation) or via an external perturbation (CME, tidal-rotation coupling). The level $k_B T_{\text{eff}}$ represents the available effective thermal energy: as long as it lies well below the barrier, ejections are rare and massive. The unstable zone ($\lambda > 0$, pink background) corresponds to the regime where the term $\lambda_3 = (\alpha - 1)U^2/R_\oplus^2$ of Eq. (10) overcomes the gravitational restoring force: it is the $\alpha > 1$ term that drives instability at large velocities.

This representation is physically powerful: ejection episodes are no longer scripted

thresholds — they are **spontaneous transitions between metastable states**, exactly as in classical statistical physics phase transitions. The 2–3 episodes emerge naturally from the potential dynamics: each ejection redistributes angular momentum, modifies Ω and ε , and recharges the barrier at a slightly different level. The physics itself generates the finite number of episodes.

5.2.3 The Central Equation: Maximum Ejectable Mass

The radial Coriolis force acting on the CMT mass must overcome effective gravity. The ejection condition gives the central equation of the theory:

$$\boxed{M_{\text{ej}}^{\text{max}} \approx \frac{2 \Omega \sin \varepsilon \cdot U_{\text{crit}} \cdot M_{\text{CMT}}}{g_{\text{eff}}(0)}} \quad (13)$$

This equation reads as follows: the faster the proto-Earth rotates (Ω large), the more favourable the obliquity ($\sin \varepsilon$ close to 1), the larger the available mass in the CMT (M_{CMT}), and the weaker the confining gravity (g_{eff} reduced by centrifugal force), the greater the ejectable mass. Each of these four parameters is independently constrained. Numerical evaluation for $T_{\text{rot}} = 5 \text{ h}$, $\varepsilon = 55^\circ$, with $M_{\text{CMT}} \approx 3.3 \times 10^{23} \text{ kg}$ and $U_{\text{crit}} \approx 10.8 \text{ km s}^{-1}$:

$$M_{\text{ej}}^{\text{max}} \approx \frac{2 \times 3.49 \times 10^{-4} \times 0.819 \times 1.0768 \times 10^4 \times 3.3 \times 10^{23}}{9.03} \approx \mathbf{2.25 \times 10^{22} \text{ kg per event.}} \quad (14)$$

5.2.4 Mass Budget: $N=2-3$ Episodes + SPA Impactor

With $f_{\text{cap}} = 0.70$ (conservative capture efficiency: the BH sheet arrives as a cohesive block, plastically welds to the PAO, with no rebound or dispersal) and $N = 2-3$:

$$M_{\text{Moon}} = \underbrace{N \cdot f_{\text{cap}} \cdot M_{\text{ej}}^{\text{max}}}_{N=2-3} + \underbrace{M_{\text{SPA}}}_{\approx 3.2 \times 10^{19} \text{ kg}} + \underbrace{\Delta M}_{\text{late additions}}. \quad (15)$$

The impactor responsible for the South Pole-Aitken basin is a differentiated body of $\approx 260 \text{ km}$ that struck at $\approx 13 \text{ km s}^{-1}$ along a north-south trajectory (Schultz et al., 2026): a documented external mass contribution that completes the budget without altering the TPT mechanism.

For $N = 2$: $M_{\text{TPT}} \approx 3.15 \times 10^{22}$ kg, compatible with $M_{\text{Moon}} = 7.34 \times 10^{22}$ kg including M_{SPA} and the documented late Hadean accretion (Zellner, 2019). For $N = 3$: $M_{\text{TPT}} \approx 4.7 \times 10^{22}$ kg. Both scenarios are physically consistent.

5.2.5 Transient Post-Ejection Precession

The ejection of $M_{\text{ej}} \approx 3\text{--}5 \times 10^{22}$ kg from the intertropical band is not a negligible event for the proto-Earth. The asymmetric mass loss instantly modifies its inertia tensor. Coupled consequences:

Modification of Ω . The ejected mass carries angular momentum $\Delta L_{\text{ej}} \approx 3.5 \times 10^{33}$ J s, giving $\Delta\Omega/\Omega \approx -12\%$ after the first episode. The proto-Earth rotates more slowly: g_{eff} , J_2 and the CMT dynamics for the next episode are all modified.

Spheroid oscillation. The reduction of Ω drives a contraction of the shape towards a rounder sphere. This rebound generates shape oscillations that perturb the CMT.

Obliquity reorganisation. The asymmetric mass loss modifies the inertia tensor: obliquity takes a new value in the range $[40^\circ, 70^\circ]$. This angle conditions the ejection direction and the accretion geometry of the next episode — the origin of the crustal dichotomy (Section 7).

Recharge window. The CMT must reorganise before recharging its energy. The estimated time, $\Delta t \sim 10^5\text{--}10^6$ yr, is short enough that the proto-PAO remains hot and deformable during the second episode.

5.3 Transition 3 — Magnetic: the Hadean Dynamo

Phase transition — Magnetic — Progressive Dynamo

The third transition is magnetic: Fe-Ni core growth reaches the threshold for compositional convection, triggering the terrestrial dynamo. This triggering, driven by the same unique motor, occurs with a predicted delay of ≈ 300 Myr after the ejections.

Core growth follows a saturation law:

$$M_{\text{core}}(t) = M_{\text{core}}^{\infty} (1 - e^{-t/\tau_{\text{seg}}}), \quad \tau_{\text{seg}} \approx 30\text{--}50 \text{ Myr}. \quad (16)$$

The dynamo triggers when the Elsasser number exceeds the threshold:

$$\Lambda_{\text{Els}} = \frac{\sigma B^2}{\rho \Omega} \gtrsim 0.1\text{--}1, \quad (17)$$

corresponding to a critical field $B_{\text{crit}} \approx 3.2 \mu\text{T}$.

Established result

Predicted delay: ≈ 300 Myr — agreement with Jack Hills zircons. Ejections cease at $t^* \approx 30\text{--}50$ Myr. The dynamo reaches B_{crit} at $t \approx 330\text{--}350$ Myr, i.e. ≈ 4.2 Ga — in agreement with Jack Hills zircon data ([Tarduno et al., 2015, 2020, 2025](#)). This delay is a quantitative prediction of the unique driver, not a fit.

6 Chronology and Observational Constraints

6.1 Proposed Chronology

Proposed chronology

From accretion to the Moon — proposed chronology.

1. $t = 0$ — **end of accretion**. Silicate mantle in near-total melting, Fe-Ni dissolved. $T_{\text{rot}} \approx 5$ h (absolute maximum). Chaotic obliquity $\varepsilon \in [40^\circ, 70^\circ]$. T-Tauri Sun, silicate atmosphere 2 000–4 000 K, intense bombardment, ^{26}Al and ^{60}Fe still active.
2. $t \approx 0\text{--}5$ **Myr**. Oblateness $J_2 \approx 0.084$ concentrates matter in the intertropical belt $|\varphi| < 30^\circ$. First transition: the CMT organises within the band.
3. **First instability event** — $t \approx$ **a few Myr**. Second transition: $\lambda > 0$, the CMT crosses the double-well barrier. Cohesive ejection beyond the Roche radius. Formation of the PAO ($R_{\text{PAO}} \approx 1\,470$ km, $a_{\text{orb}} \approx 3.51 R_\oplus$). Transient precession of the proto-Earth: $\Delta\Omega/\Omega \approx -12\%$, spheroid oscillation, obliquity reorganisation.
4. **Second episode** — $\Delta t \sim 10^5\text{--}10^6$ **yr**. Asymmetric ejection onto a still-hot and deformable proto-PAO. The asymmetry generates the **crustal dichotomy** (Section 7).
5. $t \approx 30\text{--}50$ **Myr** — **window closure**. Three convergent and irreversible mechanisms: Ω decreases (tidal braking), ε stabilises below 45° (the growing Moon stabilises obliquity), $\tau_y \rightarrow 10^6$ Pa below $T^* \approx 1\,800$ K.
6. $t \approx 330\text{--}350$ **Myr** — **third transition**. The terrestrial dynamo emerges at ≈ 4.2 Ga.

6.2 The Three Observational Constraints

The proposed mechanism satisfies the three discriminating constraints as *direct consequences* of its internal dynamics.

Constraint 1 — Isotopic similarity. The Earth-Moon isotopic identity ($\Delta^{17}\text{O} < 5$ ppm; Wiechert et al. 2001; Dauphas et al. 2014) and for Cr and Ti (Dauphas, 2017; Sossi et al., 2025) is mechanically satisfied: the lunar material comes from the contem-

poraneous Hadean terrestrial mantle at each ejection. No exotic external impactor is postulated.

Constraint 2 — Iron depletion. The lunar Fe depletion (O'Neill, 1991; Jones et al., 1986) is a direct consequence of the unique driver: at each episode, the ejected mantle is more depleted in siderophiles than the previous one. The Fe/Si gradient between lunar layers is a measurable prediction.

Constraint 3 — Angular momentum. $T_{\text{rot}}^{(0)} \approx 5 \text{ h}$ is the median value from accretion simulations (Agnor et al., 1999; Kokubo & Genda, 2010): a dynamical constraint, not a postulate.

7 Crustal Dichotomy: Natural Consequence of the Asymmetric Second Episode

GRAIL data (Wieczorek et al., 2013; Zuber et al., 2013) show that the far side of the Moon has a crust on average $\approx 20 \text{ km}$ thicker than the near side ($\bar{e}_{\text{far}} \approx 54 \text{ km}$ vs $\bar{e}_{\text{near}} \approx 34 \text{ km}$). This persistent asymmetry over 4.5 Gyr is not naturally explained by a single giant impact, nor by symmetric multiple-impact models.

The TPT scenario with $N=2-3$ predicts it **mechanically and consistently with the data under hypotheses H0–H3**. After the first episode, the proto-PAO is a body of radius:

$$R_1 = \left(\frac{3 M_1}{4\pi\rho} \right)^{1/3} \approx 1\,470 \text{ km.} \quad (18)$$

Transient post-ejection precession modifies the obliquity to a new value in $[40^\circ, 70^\circ]$. The second episode ejects a cohesive toroidal flux that does *not* uniformly envelope the proto-PAO: it reaches it along a limited solid angle (corresponding to the projected intertropical band) that becomes the **current far side**. Three mechanisms amplify the asymmetry: early gravitational locking ($\tau_{\text{lock}} \sim 10^4 \text{ yr} \ll \Delta t$), the residual obliquity of the proto-PAO ($\varepsilon_{\text{PAO}} > 45^\circ$), and the irreversible BH plastic welding.

The interface between the two accreted layers predicts a seismic signature at:

$$d \approx R_{\text{Moon}} - R_1 \approx 1\,737 - 1\,470 \approx \mathbf{267 \text{ km}}, \quad (19)$$

squarely within the seismic window 200–315 km.

8 Falsifiable Predictions

Falsifiable prediction

P1 — Main seismic interface at $d \approx 267\text{--}314\text{ km}$ (Priority 1). The N=2–3 concentric-layer structure is consistent with an acoustic impedance contrast between 200 and 315 km depth. For N=2 with masses [55%, 45%], the calculation suggests a main interface at $d = R_{\text{Moon}} - R_1 \approx 314\text{ km}$ ($R_1 \approx 1423\text{ km}$, $M_{\text{acc}}^{(1)} \approx 4.0 \times 10^{22}\text{ kg}$). The expected impedance contrast is $|R| \in [0.01, 0.04]$ depending on the inter-layer Fe/Si gradient: detectable if $|R| > 0.02$ (resolution $\leq 30\text{ km}$). If, after deployment of 3 seismic stations and detection of at least 10 earthquakes with $M > 2.5$, no phase conversion is measured in the window 200–315 km, this will constitute a **significant disagreement** with P1 and will require revision of at least one of the following parameters: the number of episodes N , the inter-layer Fe/Si impedance gradient, or the mass ratio between the first two concentric layers. Absence of contrast $|R| > 0.02$ in this window would be a strong observational argument against the theory’s most discriminating criterion. *Missions:* Chang’e 7 (August 2026), FSS, LEMS.

Falsifiable prediction

P2 — Near/far-side seismic asymmetry. The $\approx 267\text{ km}$ interface is sharper beneath the far side than beneath the near side (thicker asymmetric second episode). *Mission:* multi-station seismic network.

Falsifiable prediction

P3 — Increasing Fe/Si gradient with depth. Each successive layer originates from a more Fe-Ni-depleted mantle: the Fe/Si ratio increases from the surface inward. The SPA basin, which exposes the deep mantle, must show a higher Fe concentration than the crust. *Missions:* Chang’e 6 (far side, 2024 data), future high-resolution geochemistry.

Falsifiable prediction

P4 — Dynamo delay 290–360 Myr. The Hadean dynamo emerges ≈ 300 Myr after ejections cease. *Test:* extension of the Jack Hills zircon database ([Tarduno et al., 2025](#)).

Falsifiable prediction

P5 — Optimal ejection window $45^\circ < \varepsilon < 70^\circ$. The theory predicts that instability events occur only for $\varepsilon > 45^\circ$ (geometric threshold). The optimal range is $[50^\circ, 65^\circ]$ (radial component > 0.77). *Test:* N-body simulation of the obliquity distribution of terrestrial proto-planets.

9 Upcoming Missions: Three Validation Windows

9.1 Chang’e 7: Imminent Seismic Validation

The Chang’e 7 mission (CNSA, launch **August 2026**, Long March 5, Wenchang launch site) will land at the lunar South Pole. It carries a **dedicated seismometer**: the first seismic instrument at the South Pole since Apollo 17 (1972), and the first source of lunar seismic data independent of NASA missions.

Its data will allow direct testing of P1. If an impedance contrast $|R| > 0.02$ is measured between 200 and 315 km, the concentric stratigraphy proposed by the TPT would receive strong observational support, *prior to any full SPH simulation*. If no interface is detected under these conditions, the theory would receive a strong counter-argument on its most discriminating criterion — which would constitute an invitation to revise the assumptions on per-episode masses and the inter-layer Fe/Si gradient.

This position — a precise, quantified prediction ($d \approx 314$ km, $|R| \in [0.01, 0.04]$), an imminent test, an explicit falsification criterion — is the stance this work strives to maintain.

9.2 The South Pole-Aitken Basin: a Stratigraphic Revealer, not merely a Crater

Two distinct roles of the SPA basin in the TPT theory must be distinguished:

(a) **SPA as external mass input.** The impactor (≈ 260 km, differentiated, north-south trajectory, $\approx 13 \text{ km s}^{-1}$, [Schultz et al. 2026](#)) contributed $M_{\text{SPA}} \approx 3.2 \times 10^{19}$ kg of differentiated material to the post-TPT lunar mass. This is a documented, TPT-independent contribution.

(b) **SPA as revealer of TPT stratigraphy.** This is where the geophysical perspective is richest. The SPA impact, through its oblique north-south angle and the depth of excavation (≈ 13 km depth, ≈ 2500 km diameter), has exposed deep lunar mantle material — material which, in the TPT scenario, belongs to the innermost layer: Episode 1, derived from the most Fe-rich Hadean terrestrial mantle.

SPA did **not create** the concentric stratigraphy: it **exhumed** it. This geophysical distinction is fundamental: the layered structure pre-existed the SPA impact, encoded in TPT differentiation. The impact is merely the accidental revealer of a much older internal architecture.

Expected signatures in the data:

- **Primary mantle olivine** in uplift structures, inner rings and basin rims: signature of Episode 1 (deepest layer, richest in residual Fe-Ni);
- **Deep ejecta sheets** redistributed anisotropically along the north-south impactor trajectory;
- **Fe/Si gradient** measurable between excavated lithologies (deep mantle, Episode 1) and surface crust (Episodes 2–3, Fe-depleted).

The delicate — and acknowledged — challenge is distinguishing: primary mantle olivine, excavated cumulates, ultramafic impact melts, and shocked pyroxene-olivine mixtures. This is precisely where Artemis III will become decisive.

9.3 Artemis III: Landing on a Stratigraphic Treasure

The Artemis III mission (NASA, planned **2028–2029**) will land astronauts in the lunar South Pole region, near Shackleton crater. This landing zone is not chosen arbitrarily from the TPT perspective: recent 3D simulations ([Schultz et al., 2026](#)) demonstrate that the oblique north-south impact of the SPA impactor projected millions of tonnes of deep mantle ejecta **precisely towards the South Pole**, in a characteristic butterfly-shaped plume.

In other words: Artemis III astronauts will not need to drill kilometres deep. By bending down, they will be able to collect samples of the **deep lunar mantle** — the mantle which, in the TPT framework, is the inner layer of Episode 1, the oldest and richest in information about Hadean differentiation.

From the TPT perspective, these samples will allow direct testing of:

1. **Fe/Si composition of the innermost layer:** it must be richer in Fe than the surface crust, consistent with a Hadean mantle still only slightly depleted in siderophiles at the time of the first episode.
2. **Crystallisation age:** SPA ejecta originate from a layer accreted ≈ 4.5 Gyr ago — the age of the first TPT episode. $^{207}\text{Pb}/^{206}\text{Pb}$ dating of these rocks is a direct prediction.
3. **Isotopic signature:** identical to that of the contemporaneous terrestrial mantle of the first episode ($\Delta^{17}\text{O} < 5$ ppm), which would confirm the common mantle origin.

Established result

A remarkable convergence. Chang’e 7 (August 2026) tests the stratigraphy by seismology. Artemis III (2028–2029) tests it by direct sampling of SPA ejecta. These two independent and complementary missions converge on the same site — the lunar South Pole — and towards the same goal: probing the depths of a Moon which, according to the TPT, carries within itself, layer upon layer, the memory of the Hadean differentiation of the proto-Earth.

10 Discussion

10.1 Comparison with the Giant Impact Model

Table 2: Comparison on discriminating constraints.

Constraint	Giant impact	TPT (this work)
Isotopic similarity	Difficult: requires a very specific grazing impact	Direct mechanical consequence
Fe depletion	Requires an already-differentiated Theia	Consequence of the Fe-Ni driver
Angular momentum	Very constrained initial conditions	Median value of simulations
Crustal dichotomy	Not predicted	Predicted: post-episode-2 precession
Dynamo delay	Not predicted	Predicted: 290–360 Myr
Seismic predictions	None	P1–P3, testable by Chang’e 7

A recent review concludes: “there is currently no unambiguous geochemical or isotopic evidence for the role of an external impactor in lunar formation” (Sossi et al., 2025). This finding motivates the exploration of alternatives grounded in internal dynamics.

10.2 Acknowledged Limitations

Transparency about limitations is an epistemic requirement, not a weakness.

(1) The CMT has not yet been numerically simulated. The spontaneous organisation of a large-scale toroidal flow in a BH fluid undergoing ultra-rapid rotation is physically motivated but not demonstrated in this specific context. This is Priority 2 of the programme.

(2) $f_{\text{cap}} = 0.70$ — autonomous physical justification. Two independent mechanisms motivate this value: the geometric cohesion of the toroidal flow, which suppresses individual ballistic dispersal of ejected particles, and the mechanical integrity provided by the quench-glass crust (Section 5.1), formed in ≈ 7 min. A 30 % loss is retained as a conservative estimate, consistent with edge losses and the fraction not captured into orbit. A sensitivity analysis over $f_{\text{cap}} \in [0.50; 0.85]$ (table below) shows the theory is robust:

f_{cap}	$M_{\text{TPT}}^{(N=3)}$	fraction of M_{Moon}
0.50	$3.3 \times 10^{22} \text{ kg}$	45 %
0.70	$4.6 \times 10^{22} \text{ kg}$	63 %
0.85	$5.6 \times 10^{22} \text{ kg}$	76 %

Note. Percentages are computed relative to the reference lunar mass $M_{\text{Moon}} = 7.34 \times 10^{22} \text{ kg}$

(Wieczorek et al., 2013).

In the most unfavourable case ($f_{\text{cap}} = 0.50$), the remaining 55 % are compatible with the SPA contribution ($\approx 0.04 \%$) and documented late accretion (Zellner, 2019). This value remains to be refined by SPH simulation (Priority 2).

(3) The super-rigid profile $\alpha > 1$ is a conjecture. Physically motivated by four mechanisms, not yet demonstrated. Validation: Priority 2.

(4) The dynamo delay is an order-of-magnitude estimate. The exponential core-growth model is simplified. Estimated precision: $\pm 50 \text{ Myr}$.

10.3 Validation Programme

Priority 0 — Rheology. Experiments on yield-stress fluids in the $\text{Ma} \approx 6\text{--}8$ regime: cohesion validation (P5). Feasible in the laboratory, timeline 1–3 yr.

Priority 1 — Seismology. Chang’e 7 (August 2026), FSS, LEMS, Artemis III: detection of interfaces P1–P3. Timeline: 2026–2030.

Priority 2 — 3D SPH simulation. Emergence of the CMT on an oblate sphere in rapid rotation ($T_{\text{rot}} = 5 \text{ h}$, $\varepsilon = 55^\circ$, BH rheology). Timeline: 3–5 yr.

Priority 3 — Geochemistry. Fe/Si gradient on SPA basin ridges and the far side.

Priority 4 — Zircons. Extension of the database to refine the dynamo delay.

11 Conclusion

This work proposes that lunar formation is the product of a Triple Phase Transition in a fully molten proto-Earth rotating at ≈ 5 h, whose chaotic obliquity in the range $[40^\circ, 70^\circ]$ plays a central and hitherto underestimated geometric role.

The driver is unique and internal: progressive Fe-Ni segregation successively orchestrates the emergence of the rheological CMT in the intertropical belt $|\varphi| < 30^\circ$, its mechanical ejection in two to three massive events whose dynamics follow a double potential well, and the delayed ignition of the Hadean dynamo.

The double-well structure replaces arbitrary thresholds with spontaneous bifurcations: the episodes emerge from the physics, not from scripted conditions. Transient post-ejection precession mechanically links the episodes and generates the crustal dichotomy — a prediction that the TPT scenario alone formulates mechanically.

The falsifiability criterion is precise, the test is imminent (Chang’e 7, August 2026), the limitations are documented. Absence of impedance contrast $|R| > 0.02$ in the 200–315 km window, after 3 stations and 10 earthquakes with $M > 2.5$, would constitute a significant disagreement with P1 and would invite revision of either N or the inter-layer Fe/Si gradient: the theory commits to this explicitly. This work invites the planetological community to adjudicate on an observational and numerical basis; whatever the outcome, it places the physics of planetary differentiation and the role of axial obliquity at the heart of the problem of lunar origin.

A Adiabatic Temperature Profile (Supplement to H0)

With $T_{\text{surf}} \approx 3\,500$ K and the adiabatic gradient $dT/dr = -\alpha_{\text{th}} g T/c_p$, the central temperature reaches $\approx 5\,000$ K. The peridotite solidus at 135 GPa is $\approx 4\,300$ K (Stixrude & Lithgow-Bertelloni, 2014): melting is thermodynamically stable throughout the mantle, confirming H0.

B Dynamo Delay Estimate

With $M_{\text{core}}(t) = M_{\text{core}}^\infty (1 - e^{-t/\tau_{\text{seg}}})$, $\tau_{\text{seg}} \approx 40$ Myr and $\Lambda_{\text{Els}}^{\text{crit}} \approx 0.3$, the field exceeds $B_{\text{crit}} \approx 3.2 \mu\text{T}$ approximately 150 Myr after the end of major differentiation, i.e. $t \approx 330$ – 350 Myr corresponding to ≈ 4.2 Ga.

C Interactive Numerical Simulation

An WebGL/HTML5 simulation has been developed to illustrate the TPT dynamics in real time. It incorporates: real-time computation of λ , Ma , Λ_{ELs} and Fe-Ni segregation; silicate atmosphere 2 000–4 000 K and T-Tauri jets; visually delimited intertropical band $|\varphi| < 30^\circ$; cohesive particle flux with Roche radius crossing and plastic accretion; transient post-ejection precession; SPA impactor. Adjustable parameters: α , $\varepsilon \in [40^\circ, 70^\circ]$, Ω , T-Tauri intensity, τ_y , chaos intensity. Available as a standalone HTML file.

References

- Agnor, C. B., Canup, R. M., & Levison, H. F. 1999, *Icarus*, 142, 219–237.
- Airapetian, V. S., Glocer, A., Gronoff, G., Hébrard, É., & Danchi, W. 2016, *Nature Geoscience*, 9, 452–455.
- Aurnou, J. M., Heimpel, M., & Wicht, J. 2007, *Icarus*, 190, 110–126.
- Bertka, C. M. & Fei, Y. 1997, *Journal of Geophysical Research*, 102, 5251–5264.
- Bingham, E. C. 1922, *Fluidity and Plasticity*. New York: McGraw-Hill.
- Borg, L. E. & Carlson, R. W. 2023, *Annual Review of Earth and Planetary Sciences*, 51, 25–52.
- Bouvier, J., Matt, S. P., Mohanty, S., et al. 2014, *Protostars and Planets VI*, eds. H. Beuther et al. Tucson: University of Arizona Press, 433–450.
- Briaud, A., Ganino, C., Fienga, A., Mémin, A., & Rambaux, N. 2023, *Nature*, 617, 743–746.
- Busse, F. H. 1970, *Journal of Fluid Mechanics*, 44, 441–460.
- Cameron, A. G. W. & Ward, W. R. 1976, *Lunar and Planetary Science Conference*, 7, 120–122.
- Canup, R. M. 2004, *Icarus*, 168, 433–456.
- Canup, R. M. 2012, *Science*, 338, 1052–1055.
- Caricchi, L., Burlini, L., Ulmer, P., et al. 2007, *Earth and Planetary Science Letters*,

- 264, 402–419.
- Chandrasekhar, S. 1969, *Ellipsoidal Figures of Equilibrium*. New Haven: Yale University Press.
- Chambers, J. E. 2001, *Icarus*, 152, 205–224.
- Chambers, J. E. 2013, *Icarus*, 224, 43–56.
- Che, X., Nemchin, A., Liu, D., et al. 2021, *Science*, 374, 887–890.
- Christensen, U. R. & Aubert, J. 2006, *Geophysical Journal International*, 166, 97–114.
- Čuk, M. & Stewart, S. T. 2012, *Science*, 338, 1047–1052.
- Dauphas, N., et al. 2014, *Philosophical Transactions of the Royal Society A*, 372, 20130244.
- Dauphas, N. 2017, *Nature*, 541, 521–524.
- Dauphas, N., Zhang, Z. J., Chen, X., et al. 2025, *Proceedings of the National Academy of Sciences*, 122, e2413802121.
- Deitrick, R., Barnes, R., Bitz, C., et al. 2022, *Astronomical Journal*, 163, 179.
- Drilleau, M., et al. 2025, *EPSC-DPS Joint Meeting 2025*, Abstract EPSC-DPS2025-1524.
- Druitt, T. H. 1992, *Geological Society Special Publications*, 145, 145–182.
- Dziewonski, A. M. & Anderson, D. L. 1981, *Physics of the Earth and Planetary Interiors*, 25, 297–356.
- Elkins-Tanton, L. T., et al. 2011, *Earth and Planetary Science Letters*, 304, 326–336.
- Elkins-Tanton, L. T. 2008, *Earth and Planetary Science Letters*, 271, 181–191.
- Elkins-Tanton, L. T. 2012, *Annual Review of Earth and Planetary Sciences*, 40, 113–139.
- Feigelson, E. D. & Montmerle, T. 1999, *Annual Review of Astronomy and Astrophysics*, 37, 363–408.
- Flasar, F. M. & Birch, F. 1973, *Journal of Geophysical Research*, 78, 6101–6114.
- Garcia, R. F., Khan, A., Drilleau, M., et al. 2019, *Space Science Reviews*, 215, 50.
- Gastine, T., Wicht, J., & Aurnou, J. M. 2014, *Earth and Planetary Science Letters*,

- 408, 370–380.
- Giordano, D., Russell, J. K., & Dingwell, D. B. 2008, *Earth and Planetary Science Letters*, 271, 123–134.
- Hartmann, W. K. & Davis, D. R. 1975, *Icarus*, 24, 504–515.
- Hekinian, R., et al. 1989, *Earth-Science Reviews*, 25, 213–251.
- Herschel, W. H. & Bulkley, R. 1926, *Kolloid-Zeitschrift*, 39, 291–300.
- Høiland, E. 1941, *Avhandlingar Norske Videnskaps-Akademi Oslo*, 11, 1–24.
- Huss, G. R., et al. 2009, *Geochimica et Cosmochimica Acta*, 73, 4922–4945.
- Johansen, A. & Lambrechts, M. 2017, *Annual Review of Earth and Planetary Sciences*, 45, 359–387.
- Jones, J. H., Delano, J. W., & Longhi, J. 1986, *Journal of Geophysical Research*, 91, D365–D374.
- Königl, A. 1991, *Astrophysical Journal Letters*, 370, L39–L43.
- Kokubo, E. & Genda, H. 2010, *Astrophysical Journal Letters*, 714, L21–L25.
- Laskar, J., Joutel, F., & Robutel, P. 1993, *Nature*, 361, 615–617.
- Laskar, J. 1993, *Physica D*, 67, 257–281.
- Lock, S. J. & Stewart, S. T. 2017, *Journal of Geophysical Research: Planets*, 122, 950–982.
- Mader, H. M., Llewellyn, E. W., & Mueller, S. P. 2013, *Journal of Volcanology and Geothermal Research*, 257, 135–166.
- McLachlan, N. W. 1947, *Theory and Application of Mathieu Functions*. Oxford: Clarendon Press.
- Melosh, H. J. 1990, *Annual Review of Earth and Planetary Sciences*, 18, 1–24.
- Monaghan, J. J. 2005, *Reports on Progress in Physics*, 68, 1703–1759.
- Nguyen, T. G., Cowan, N. B., Banerjee, A., & Spiegelman, M. 2020, *Monthly Notices of the Royal Astronomical Society*, 499, 4977–4993.
- Nomura, R., Ozawa, H., Tateno, S., et al. 2011, *Nature*, 473, 199–202.
- O’Brien, D. P., Morbidelli, A., & Levison, H. F. 2006, *Icarus*, 184, 39–58.

- O'Neill, H. S. C. 1991, *Geochimica et Cosmochimica Acta*, 55, 1135–1157.
- O'Neill, C. & Debaille, V. 2014, *Earth and Planetary Science Letters*, 406, 49–58.
- Panning, M. P., et al. 2022, *Farside Seismic Suite: Lunar Seismology in the Artemis Era*, AGU Fall Meeting Abstract.
- Porco, C. C., et al. 2006, *Science*, 311, 1393–1401.
- Pozzo, M., Davies, C., Gubbins, D., & Alfè, D. 2012, *Nature*, 485, 355–358.
- Price, D. J. 2012, *Journal of Computational Physics*, 231, 759–794.
- Rayleigh, Lord 1917, *Proceedings of the Royal Society A*, 93, 148–154.
- Raymond, S. N., O'Brien, D. P., Morbidelli, A., & Kaib, N. A. 2014, *Icarus*, 203, 644–662.
- Read, P. L., et al. 2018, *Nature Astronomy*, 2, 211–219.
- Rhines, P. B. 1975, *Journal of Fluid Mechanics*, 69, 417–443.
- Roche, É. A. 1849, *Académie des Sciences et Lettres de Montpellier*, 1, 243–262.
- Rubie, D. C., Melosh, H. J., Reid, J. E., et al. 2003, *Earth and Planetary Science Letters*, 205, 239–255.
- Rubie, D. C., Jacobson, S. A., Morbidelli, A., et al. 2015, *Icarus*, 248, 89–108.
- Rufu, R., Aharonson, O. & Perets, H. B. 2017, *Nature Geoscience*, 10, 89–94.
- Safronov, V. S. 1969, *Evolution of the Protoplanetary Cloud and Formation of the Earth and Planets* (English transl. 1972, NASA TT F-677). Moscow: Nauka.
- Schultz, P. H., et al. 2026, *Science Advances*, in press.
- Shu, F. H., Najita, J., Ostriker, E., et al. 1994, *Astrophysical Journal*, 429, 781–796.
- Solomatov, V. S. 2000, in *Origin of the Earth and Moon*, eds. R. M. Canup & K. Righter, Tucson: University of Arizona Press, 323–338.
- Sossi, P. A., et al. 2025, *Treatise on Geochemistry*, 3rd ed., 417–479.
- Stixrude, L. & Lithgow-Bertelloni, C. 2014, *Philosophical Transactions of the Royal Society A*, 372, 20130076.
- Sukoriansky, S., Dikovskaya, N., & Galperin, B. 2007, *Journal of the Atmospheric Sciences*, 64, 3312–3327.

- Tarduno, J. A., Cottrell, R. D., Davis, W. J., et al. 2015, *Science*, 349, 521–524.
- Tarduno, J. A., Cottrell, R. D., Bono, R. K., et al. 2020, *Proceedings of the National Academy of Sciences*, 117, 2309–2318.
- Tarduno, J. A., Zhou, T., Huang, W., & Jodder, J. 2025, *National Science Review*, 12, nwaf082.
- Tsiganis, K., Gomes, R., Morbidelli, A. & Levison, H. F. 2005, *Nature*, 435, 459–461.
- Urey, H. C. 1955, *Proceedings of the National Academy of Sciences*, 41, 127–144.
- Valley, J. W., Peck, W. H., King, E. M., & Wilde, S. A. 2002, *Geology*, 30, 351–354.
- van Dijk, M. R., Nicholls, H. & Lichtenberg, T. 2026, *Planetary Science Journal*, submitted (arXiv:2511.00952).
- Walsh, K. J., et al. 2011, *Nature*, 475, 206–209.
- Weber, R. C., Lin, P.-Y., Garnero, E. J., Williams, Q., & Lognonné, P. 2011, *Science*, 331, 309–312.
- Wetherill, G. W. 1990, *Annual Review of Earth and Planetary Sciences*, 18, 205–256.
- Wiechert, U., Halliday, A. N., Lee, D.-C., et al. 2001, *Science*, 294, 345–348.
- Wieczorek, M. A., et al. 2013, *Science*, 339, 671–675.
- Wilde, S. A., Valley, J. W., Peck, W. H., & Graham, C. M. 2001, *Nature*, 409, 175–178.
- Yin, Q., Jacobsen, S. B., Yamashita, K., et al. 2002, *Nature*, 418, 949–952.
- Zellner, N. E. B. 2019, *Origins of Life and Evolution of Biospheres*, 49, 261–274.
- Zuber, M. T., et al. 2013, *Science*, 339, 668–671.
- Balmforth, N. J. & Craster, R. V. 2001, *Journal of Fluid Mechanics*, 426, 297–323.
- Frigaard, I. A. & Nouar, C. 2017, in *Viscoplastic Fluids: From Theory to Application*. Springer, 201–238.